

```
/* C Program to sort an array in ascending order using Insertion Sort */
```

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    int n, i, j, temp;
```

```
    int arr[64];
```

```
    printf("Enter number of elements\n");
```

```
    scanf("%d", &n);
```

```
    printf("Enter %d integers\n", n);
```

```
    for (i = 0; i < n; i++)
```

```
    {
```

```
        scanf("%d", &arr[i]);
```

```
    }
```

```
    for (i = 1; i < n; i++)
```

```
    {
```

```
        j = i;
```

```
        while (j > 0 && arr[j - 1] > arr[j])
```

```
        {
```

```
            temp = arr[j];
```

```
            arr[j] = arr[j - 1];
```

```
            arr[j - 1] = temp;
```

```
            j--;
```

```
        }
```

```
}  
  
printf("Sorted list in ascending order:\n");  
  
for (i = 0; i < n; i++)  
{  
    printf("%d\n", arr[i]);  
}  
  
return 0;  
}
```